

JEE MAIN REVISION MASTER: THERMODYNAMICS

2026 3 QUESTIONS 5% WEIGHTAGE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 14 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Enthalpy, Hess's Law & Bond Energies	8 PYQs	Thermochemistry
Moderate	Entropy & Gibbs Free Energy Spontaneity	7 PYQs	Spontaneity

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Enthalpy, Hess's Law & Bond Energies	$\Delta H = \Delta U + \Delta n_g RT, \quad \Delta_r H^\circ = \sum \Delta_f H^\circ(\text{products}) - \sum \Delta_f H^\circ(\text{reactants})$
Entropy & Gibbs Free Energy Spontaneity	$\Delta G = \Delta H - T\Delta S, \quad \Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} > 0, \quad \Delta G^\circ = -RT \ln K$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Enthalpy, Hess's Law & Bond Energies (High Yield — 8 PYQs)

Core Principle: State functions, standard enthalpy of formation, Hess's law of constant heat summation, and bond energy calculations.

■ **Caution / JEE Pitfall:** Standard enthalpy of formation $\Delta_f H^\circ$ of pure elements in their standard reference state is strictly ZERO (e.g. $O_2(g)$, $C_{\text{(graphite)}}$).

■ **Speed Trick:** When using bond enthalpies: $\Delta H_{\text{(reaction)}} = \sum BE(\text{bonds broken in reactants}) - \sum BE(\text{bonds formed in products})$.

Representative Source PYQs:

- [2025 Shift] Q#1 (MCQ): "Which one of the following reactions has $\Delta H = \Delta U$?..." [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "Identify the incorrect statements among the following: (a) All enthalpies of fusion are positive (b) The magnitude of enthalpy change does not..." [Ans: N/A]
- [2025 Shift] Q#4 (MCQ): "Given below are two statements. Statement-I : Adiabatic work done is positive when work is done on the system and internal energy of the sys..." [Ans: N/A]

3.2 Entropy & Gibbs Free Energy Spontaneity (Moderate Yield — 7 PYQs)

Core Principle: Second and third laws of thermodynamics, entropy changes in phase transitions, and Gibbs criteria for spontaneous change.

■ **Caution / JEE Pitfall:** For an endothermic reaction ($\Delta H > 0$) with $\Delta S > 0$, the reaction is spontaneous ONLY at HIGH temperatures where $T > (\Delta H / \Delta S)$.

■ **Speed Trick:** At equilibrium: $\Delta G = 0$ implies $T_{\text{(equilibrium)}} = (\Delta H / \Delta S)$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "A gas can be taken from A to B via two different paths ACB and ADB. When path ACB is used, 60 J of heat flows into the system and 30 J o..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "Which of the following is a correct statement for a thermodynamic system?..." [Ans: N/A]
- [2026 Shift] Q#3 (MCQ): "A: Entropy of a perfect crystalline solid at absolute zero approaches zero. B: For spontaneity of a reaction, $T\Delta S > \Delta H$. Among t..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

A gas can be taken from A to B via two different paths ACB and ADB. When path ACB is used, 60 J of heat flows into the system and 30 J of work is done by the system. If path ADB is used, work done by the system is 10 J. The heat flow into the system in path ADB is

- (A) 80 J
- (B) 20 J
- (C) 100 J
- (D) 40 J

Correct Answer: None

Solution Analysis: By the first law of thermodynamics, $\Delta U = Q - W$. Here, Q is the heat supplied to the system, and W is the work done by the system. For path ACB: Heat added, $Q_{\text{(ACB)}} = 60$ J Work done, $W_{\text{(ACB)}} = 30$ J So, $\Delta U = 60 - 30 = 30$ J The change in internal energy depends only on the initial state A and final state B. It does not depend on the path followed. So, for path ADB also, $\Delta U = 30$ J For path ADB: Work done, $W_{\text{(ADB)}} = 10$ J Using the first law again, $\Delta U = Q_{\text{(ADB)}} - W_{\text{(ADB)}}$ $30 = Q_{\text{(ADB)}} - 10$ $Q_{\text{(ADB)}} = 30 + 10 = 40$ J The heat flow into the system in path ADB is 40 J.

Benchmark Problem 2: KCET 2025 (Q#1)

Which one of the following reactions has $\Delta H = \Delta U$?

- (A) $\text{CaCO}_3(\text{s}) \rightarrow \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$
- (B) $\text{C}_6\text{H}_6(\text{l}) + (15/2)\text{O}_2(\text{g}) \rightarrow 6\text{CO}_2(\text{g}) + 3\text{H}_2\text{O}(\text{l})$
- (C) $2\text{HI}(\text{g}) \rightleftharpoons \text{H}_2(\text{g}) + \text{I}_2(\text{g})$
- (D) $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$

Correct Answer: None

Solution Analysis: To determine which reaction has $\Delta H = \Delta U$, we use the equation: $\Delta H = \Delta U + \Delta n_{\text{(g)}} \cdot RT$ Where: ΔH is the change in enthalpy. ΔU is the change in internal energy. $\Delta n_{\text{(g)}}$ is the change in the number of moles of gas. R is the ideal gas constant. T is the temperature in Kelvin. For the reaction: $2\text{HI}(\text{g}) \rightleftharpoons \text{H}_2(\text{g}) + \text{I}_2(\text{g})$ The change in the number of moles of gas ($\Delta n_{\text{(g)}}$) can be calculated as follows: On the reactant side, there are 2 moles of gas. On the product side,